The leftward motion of arm *AB* decreases magnetic flux. The induced current, this time, produces **a magnetic field in the inward direction (red-colored crosses) i.e., in the direction of the existing magnetic field** (Figure 4.6(c)). Therefore, the flux decrease is opposed by the flow of induced current. From this, it is found that induced current flows in clockwise direction.

Illustration 2

Let us move a bar magnet towards the solenoid, with its north pole pointing the solenoid as shown in Figure 4.7(b). This motion increases the magnetic flux of the coil which in turn, induces an electric current. Due to the flow of induced current, the coil becomes a magnetic dipole whose two magnetic poles are on either end of the coil.

In this case, the cause producing the induced current is the movement of the magnet. According to Lenz's law, the induced current should flow in such a way that it opposes the movement of the north pole towards coil. It is possible if the end nearer to the magnet becomes north pole (Figure 4.7(b)). Then it repels the north pole of the bar magnet and opposes the movement of the magnet. Once pole ends are known, the direction of the induced current could be found by using right hand thumb rule.

When the bar magnet is withdrawn, the nearer end becomes south pole which attracts north pole of the bar magnet, opposing the receding motion of the magnet (Figure 4.7(c)).

Thus the direction of the induced current can be found from Lenz's law.

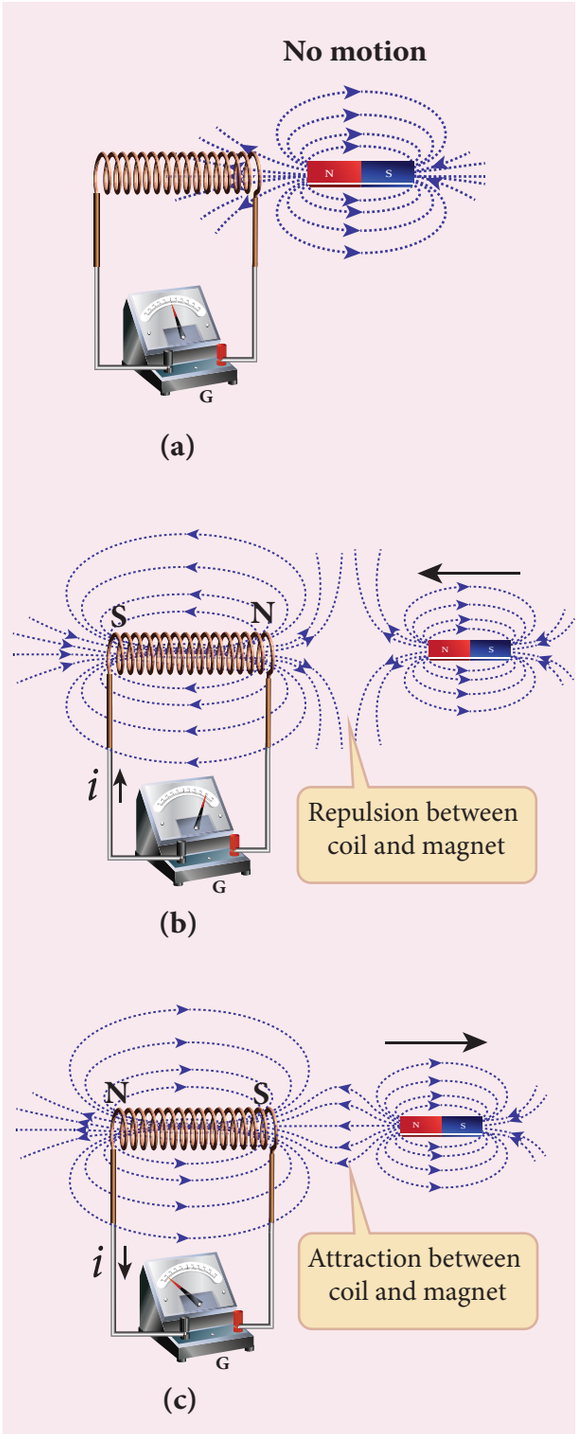


Figure 4.7 Second illustration of Lenz's law

Conservation of energy

The truth of Lenz's law can be established on the basis of the law of conservation of energy. The explanation is as follows: According to Lenz's law, when a magnet is moved either towards or away from a coil, the induced current produced opposes its motion. As a result, there will always be a

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